

James V. Raj

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EDUCATION

University of California, Los Angeles

Expected June 2027

Ph.D. in Physics. Advisor: Mason A. Porter. Advanced to candidacy March 2025.

California Polytechnic State University, San Luis Obispo

2021

B.S. in Physics

RESEARCH INTERESTS

Physical learning with local learning rules; complex systems; statistical physics; network science; topological data analysis and persistent homology; networks that arise from physical systems, including scale-rich metamaterials and hyperuniform point patterns.

PUBLICATIONS

Peer-reviewed journal articles

R. Sengupta, M. D. Tikekar, **J. V. Raj**, K. T. Delaney, M. C. Villet, and G. H. Fredrickson, "Phase-field simulations of morphology development in reactive polymer blending," *J. Rheol.* **67**, 1 (2023). [doi:10.1122/8.0000523](https://doi.org/10.1122/8.0000523)

Preprints and papers under review

J. V. Raj, X. Sun, C. E. Maher, K. A. Newhall, and M. A. Porter, “Local geometric and transport properties of networks that are generated from hyperuniform point patterns,” [arXiv:2511.21082](#) (2025); in revision at *Phys. Rev. E*. Code: [github.com/DMREF-networks/HuPPI-Network-Analysis](#).

Manuscripts in preparation

J. V. Raj, L. Gyllingberg, P. B. Pietrzkowski, M. Stern, and M. A. Porter, “Network structure controls how quickly local learning opens spectral gaps in spring networks” (2026).

J. V. Raj, C. E. Maher, K. A. Newhall, and M. A. Porter, “Size-resolved H_0 persistence as an exact coalescence observable” (2026).

J. V. Raj, C. E. Maher, K. A. Newhall, and M. A. Porter, “A directional transport surrogate for stiffness in disordered beam networks” (2026).

HONORS, AWARDS, AND FUNDING

Outstanding Teaching Award, Department of Physics and Astronomy, UCLA, for 2022 to 2024
the 2022 to 2023 and 2023 to 2024 academic years

Graduate research support, NSF DMREF award CMMI-2323343 (UCLA) 2024 to present

Future Leaders in Advanced Materials Internship Award 2020

William and Linda Frost Undergraduate Research Award, Cal Poly 2019

PRESENTATIONS

Contributed talks

“Network structure shapes accessible solutions in physical learning” October 2026

Conference on Complex Systems (CCS 2026), Binghamton University, Binghamton, NY

“Topological and transport properties of spatially embedded networks generated by hyperuniform point patterns” March 2025
APS March Meeting, Anaheim, CA

“Understanding how the glass transition concentration affects polymer membrane microstructure” August 2020
Future Leaders in Advanced Materials Summer Symposium

Posters

“Electrical current characteristics of simulated carbon nanotube network field-effect transistors” October 2019
APS Far West Section Meeting

RESEARCH EXPERIENCE

Graduate Student Researcher, UCLA 2024 to present
Advisor: Mason A. Porter, Department of Mathematics

Graduate Student Researcher, UCLA 2022 to 2023
Advisor: Shenshen Wang, Department of Physics and Astronomy
Theoretical biophysics of adaptive immunity and evolving populations.

Postbaccalaureate Researcher, Materials Research Laboratory, UC Santa Barbara 2021 to 2022
Advisor: Glenn H. Fredrickson, Department of Chemical Engineering
Designed and ran phase-field simulations of reactive polymer blending under shear flow on UCSB’s high-performance computing clusters; wrote the Python simulation-input and ParaView analysis scripts.

Summer Research Intern (NSF-funded), UC Santa Barbara 2020
Advisor: Glenn H. Fredrickson
Simulated polymer membrane formation on high-performance computing clusters; wrote Python tools to quantify pore structure in SEM images and compare it with simulations.

Soft Matter Research Assistant, Cal Poly 2019 to 2021
Advisor: Hilary Jacks
Statistical analysis of memory formation in non-Brownian suspensions under cyclic shear, including an algorithm to read multiple stored memories from particle positions.

Nanoelectronics Research Assistant, Cal Poly 2018 to 2019
Advisor: Colleen Marlow; includes a 2019 summer research internship
Simulations of carbon nanotube network field-effect transistors: network morphology, percolation, and sensitivity to electrostatic gating.

TEACHING

Teaching Assistant, Department of Physics and Astronomy, UCLA 2023 to 2025
Upper division: Introduction to Galactic Astronomy (Winter 2023, Winter 2024); Thermal Physics (Spring 2023); Mathematical Methods (Fall 2023)
Lower division: Mechanics (Winter 2025); Oscillations, Waves, Electric and Magnetic Fields (Fall 2024, Spring 2025)

MENTORING

Mentor to eight undergraduate students, UCLA Marginalized Identities in Physics & Astronomy (MiPA) 2023 to 2025

Mentor to two graduate students 2023 to 2025

SERVICE

Assisted with a referee report for *Interfaces and Free Boundaries* 2025

Administrator of the Porter research group's shared Linux compute server 2025 to present

Vice President, Sigma Pi Sigma (physics honor society), Cal Poly 2020 to 2021

COMPUTATIONAL SKILLS

Languages: Python, Bash, $\text{\LaTeX}$, JavaScript/HTML/CSS, Rust (numerical kernels called from Python)

Scientific computing: NumPy, SciPy (sparse linear algebra, eigensolvers), NetworkX, pandas, Matplotlib, Plotly, ParaView; parallel computing with multiprocessing and joblib; GUDHI, Ripser, and persim for persistent homology

Machine learning: PyTorch (CUDA, `torch.compile`, differentiable optimization); LoRA fine-tuning of diffusion models; local deployment of open-weight LLMs with llama.cpp (quantization, speculative decoding, mixture-of-experts offloading)

Systems: batch jobs on high-performance computing clusters; Linux server administration (user accounts, systemd services, networking) for a 96-core, 1 TB RAM GPU server; Git and GitHub; pytest